

Data Types

Imagine a bank account.

A bank account contains different kinds of information:

- Account Holder Name → "Shanmuka"
- Account Balance → 50000.75
- Account Active → True
- Transaction IDs → [101, 102, 103] → Collection

All these values are different.

Python needs to know:

- Which value is text?
- Which value is a number?
- Which value is a yes/no answer?
- Which value is a collection?

This is where **Data Types** become important.

variables
↑
What kind of data it stores →

Variables:-

name = "Sophie" ✓
age = 16 ✓
wt = 45.75 ✓

Data Types

Single Value

Collection Data Types

These store one value at a time.

Examples:

- int ✓ → age → 12
- float ✓ → wt → 42.35
- str ✓ → name → "Tejas"
- bool ✓ → isactive → false

that can change

These store multiple values together.

Examples:

- list
- tuple
- set
- dict

you can access by Index

Data Type	Example	Mutable/Immutable	Ordered/Unordered	Real-Life Example
int	100	Immutable	N/A	Roll Number
float	99.5	Immutable	N/A	Product Price
str	"Python"	Immutable	Ordered	Student Name
bool	True	Immutable	N/A	Login Status
list	[1,2,3]	Mutable	Ordered	Shopping Cart
tuple	(1,2,3)	Immutable	Ordered	Coordinates
set	{1,2,3}	Mutable	Unordered	Unique IDs
dict	{"name": "Rahul"}	Mutable	Insertion-Ordered	Student Record

single

collection

List ← collection data types

➤ List is a group of **similar** or **different** data types.

Ex- marks = [14, 15, 8, 9, 11]

random_hobbies_age = [volleyball, 24, TRUE]

g know python

int

boolean.

nested list = [[1, 2, 3]]

A List allows us to store multiple values in a single variable.

Lists are:

✓ Ordered

✓ Mutable

✓ Allow Duplicate Values

→ we can access it using index

→ It can be change

```
subjects = ["Math", "Science", "English"]
```

```
# Print the entire list of subjects
```

```
print(subjects)
```

```
# Print the data type of the 'subjects' variable
```

```
print(type(subjects))
```

Accessing a list

```
#~ Accessing a List using index
```

```
subjects = ["Math", "Science", "English"]
```

```
#      0      1      2
```

```
print(subjects[0])
```

```
print(subjects[1])
```

```
print(subjects[2])
```

Modifying a List

```
subjects = ["Math", "Science", "English"]  
#         0      1      2
```

```
subjects[1] = "Physics"  
print(subjects)
```